Mathematical Methods

Derivative-Many Variables

Delhi School of Economics

Derivative-Many Variables

Readings

Simon and Blume: Chapter 14

Partial Derivative

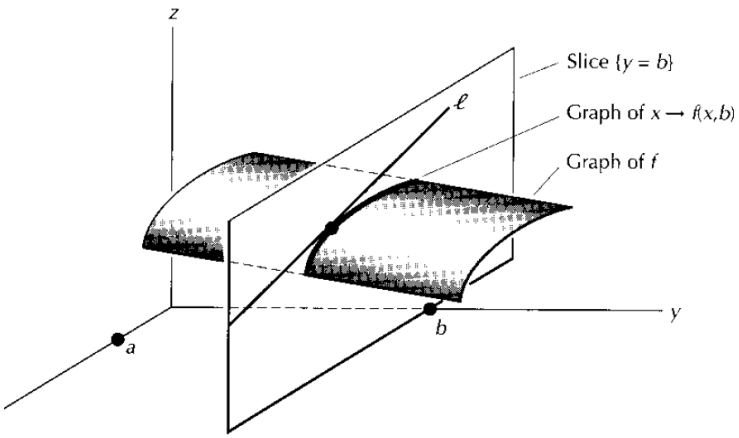
Partial derivative of $f(x_1, x_2, \ldots, x_L)$ at $\bar{x} = (\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_L)$, with respect to x_i is defined as

$$\frac{\partial f}{\partial x_i}(\bar{x}) = \lim_{t \to 0} \frac{f(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_i + t, \ldots, \bar{x}_L) - f(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_i, \ldots, \bar{x}_L)}{t}$$

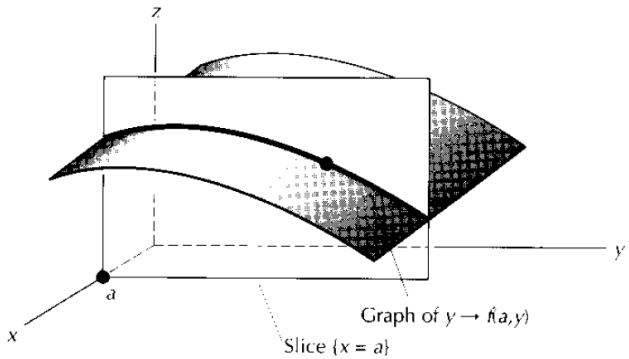
$$= \lim_{t \to 0} \frac{f(\bar{x} + te_i) - f(\bar{x})}{t}$$

Captures the 'rate of change in output' with respect to x_i.

Partial Derivative



The graph of $x \longmapsto f(x, b)$ on the slice $\{y = b\}$.



The graph of $y \longmapsto f(a, y)$ on the slice $\{x = a\}$.

Example: $f(x_1, x_2) = x_1 {x_2}^2$.
Partials of f at $\bar{x}$ is $\frac{\partial f}{\partial x_1}(\bar{x}) = \bar{x}_2^2$, $\frac{\partial f}{\partial x_2}(\bar{x}) = 2\bar{x}_1\bar{x}_2$

Diffrentiability: Linear Approximation

f is **differentiable** at $\bar{x}$ if there is a $(1 \times L)$ vector γ such that

$$\lim_{y \to \bar{x}} \frac{[f(y) - f(\bar{x}) - \gamma \cdot (y - \bar{x})]}{||y - \bar{x}||} = 0$$

That is for all sequences $\{y_n\}_{n=1}^{\infty}$ converging to $\bar{x}$,

$$\left\{ \frac{[f(y_n) - f(\bar{x}) - \gamma \cdot (y_n - \bar{x})]}{||y_n - \bar{x}||} \right\}_{n=1}^{\infty}$$

sequence converges to 0.

γ is called the **gradient** of f at $\bar{x}$. Gradient is denoted by $\nabla f(\bar{x})$.

Example

- $f : \mathcal{R}^2 \to \mathcal{R}$,

$$f(x,y) = \begin{cases} \dfrac{xy}{\sqrt{x^2+y^2}} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

  f is continuous at $(0,0)$ but it is not differentiable at $(0,0)$.

- $f : \mathcal{R}^2 \to \mathcal{R}$. $f(x,y) = xy$ is continuous and differentiable at every point in $\mathcal{R}^2$

Differentiability and Continuity

Result If f is differentiable at $\bar{x}$ then f is continuous at $\bar{x}$.

Proof: Since f is differentiable at $\bar{x}$, for all $y \neq \bar{x}$, we can write

$$f(y) - f(\bar{x}) = \frac{[f(y) - f(\bar{x}) - \nabla f(\bar{x}) \cdot (y - \bar{x})]}{||y - \bar{x}||} ||y - \bar{x}|| + \nabla f(\bar{x}) \cdot (y - \bar{x})$$

Taking limit as $y \to \bar{x}$, we get

$$\lim_{y \to \bar{x}} [f(y) - f(\bar{x})] = 0$$

Gradient and Partial Derivatives

Result: If f is differentiable at $\bar{x}$ then partial derivates of f also exist at $\bar{x}$. Moreover

$$\nabla f(\bar{x}) = \left(\frac{\partial f}{\partial x_1}(\bar{x}), \frac{\partial f}{\partial x_2}(\bar{x}), \ldots, \frac{\partial f}{\partial x_L}(\bar{x}) \right)$$

Proof: Take any sequence $\{t_n\}_{n=1}^{\infty}$ converges to 0. Define $y_n = \bar{x} + t_n e_i$. Note that $\{y_n\}_{n=1}^{\infty}$ converges to $\bar{x}$.

Hence

$$\lim_{t \to 0} \frac{[f(\bar{x} + te_i) - f(\bar{x}) - \gamma \cdot (te_i)]}{||te_i||} = 0$$

$$\Rightarrow \lim_{t \to 0} \frac{f(\bar{x} + te_i) - f(\bar{x})}{t} = \gamma \cdot (e_i)$$

$$\Rightarrow \frac{\partial f}{\partial x_i}(\bar{x}) = \gamma_i$$

Linear Approximation

- $f(\bar{x}+h)$ is approximated by $[f(\bar{x})+\nabla f(\bar{x})\cdot h]$. That is

$$f(\bar{x}+h)=f(\bar{x})+\sum_{i=1}^{L}\frac{\partial f}{\partial x_i}(\bar{x})h_i$$

- Equivalently $[f(\bar{x}+h)-f(\bar{x})]$ is approximated by

$$\sum_{i=1}^{L}\frac{\partial f}{\partial x_i}(\bar{x})h_i$$

which is called **total derivative**.

- The error term is denoted by $e(h)$,

$$\lim_{h\to 0}\frac{e(h)}{||h||}=0$$

Gradient and Partial Derivatives

Result (without proof)

If partial derivatives of f exist and these are continuous functions in some neighbourhood $B_\epsilon(\bar{x})$ around $\bar{x}$, then f is differentiable at $\bar{x}$.

Tangent Plane

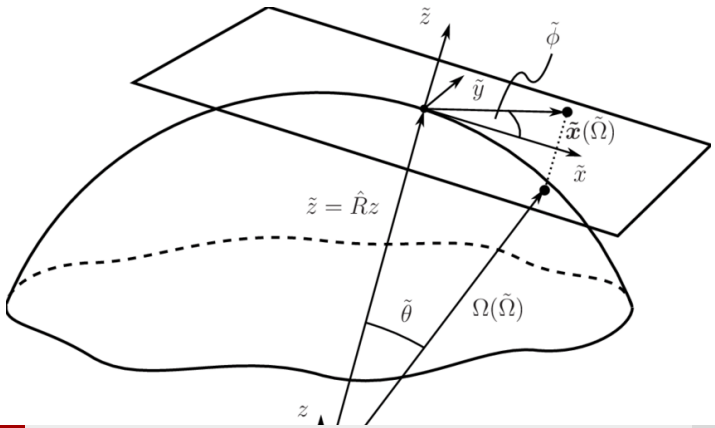
Consider the tangent lines corresponding to partial derivatives. There are L such tangent lines as follows. For $i = 1, 2, \ldots, L$

$$f(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_i + t, \ldots, \bar{x}_L) = f(\bar{x}) + \frac{\partial f}{\partial x_i}(\bar{x})t$$

There is a unique plane containing all these straight lines. This is called the **gradient plane**. Equation of this line is given by

$$z - f(\bar{x}) = \sum_{i=1}^{L} \frac{\partial f}{\partial x_i}(\bar{x})(x_i - \bar{x}_i)$$

Diagram of Tangent plane:



Directional Derivative

Partial derivatives told us 'rate of change' in the functional value along the axes.

Let us now fix any $u \in \mathcal{R}^L$. What is the 'rate of change' in the functional value in the direction of u?

Directional derivative of f at $\bar{x}$ in the direction of u is defined as

$$D_u f(\bar{x}) = \lim_{t \to 0} \frac{f(\bar{x} + tu) - f(\bar{x})}{t}$$

When f is differentiable, this is by definition

$$D_u f(\bar{x}) = \nabla f(\bar{x}) \cdot u$$

Relation between Contour and Gradient

Result The gradient is orthogonal to contour and gives the direction of steepest increase

Intuition: This proof is for $L = 2$ (general proof requires 'Implicit function theorem'). By the directional derivative expression,

$$D_u f(\bar{x}) = \nabla f(\bar{x}) \cdot u = ||\nabla f(\bar{x})|| \cdot ||u|| \cos\theta$$

where θ is the angle between $\nabla f(\bar{x})$ and u. Thus the direction maximizing $D_u f(\bar{x})$ must have $\theta = 0$. Hence the gradient is the direction of steepest increase.

On the other hand, the direction in which f does not change is given by the tangent of the contour. This must be the direction along which direction derivative is 0. Hence $\theta = \frac{\pi}{2}$. That is, gradient is orthogonal to contour.

Higher Order Derivatives

- **Second order partial derivative** of f with respect to x_i and x_j at $\bar{x}$ is defined as

$$\left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right)(\bar{x}) = \left(\frac{\partial}{\partial x_i}\left(\frac{\partial f}{\partial x_j}\right)\right)(\bar{x})$$

- Second order partial derivative matrix is also called **Hessian matrix**

$$D^2 f(\bar{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_l} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_l} \\ \cdots & \cdots & \cdots & \cdots \\ \frac{\partial^2 f}{\partial x_l \partial x_1} & \frac{\partial^2 f}{\partial x_l \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_l \partial x_l} \end{bmatrix}$$

Higher Order Derivatives

Result: (without proof)

If first and second order partial derivative of f exists for all for all points in the domain and these partial derivatives are continuous functions in the domain then

$$\left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right)(\bar{x}) = \left(\frac{\partial^2 f}{\partial x_j \partial x_i}\right)(\bar{x})$$

That is $D^2 f(\bar{x})$ is symmetric matrix.

Chain Rule

Sometime we shall deal with situations where inputs $x_1, x_2, \ldots, x_L$ are functions of a parameter $t \in \mathcal{R}$. Then we can write a composite function $g(t) = f(x_1(t), \ldots, x_L(t))$, where $g : \mathcal{R} \to \mathcal{R}$.

We may want to know how g changes with t.

Chain rule:
(i) f is continuously differentiable (differentiable and partials are continuous) on a neighbourhood around
$x(t_0) = \big(x_1(t_0), x_2(t_0), \ldots x_L(t_0)\big)$.

(ii) For each k, $x_k(t)$ is continuously differentiable (differentiable and derivative is continuous) on an interval around t_0.

Then

$$g'(t_0) = \nabla f(x(t_0)) \cdot x'(t_0)$$

where $x'(t_0) = (x_1'(t_0), x_2'(t_0), \ldots, x_L'(t_0))$

Proof: Chain Rule (Optional)

$$f(y) = f(z) + \sum_{i=1}^{L} \frac{\partial f}{\partial x_i}(z)(y_i - z_i) + e(y - z)$$

where,

$$\lim_{y \to z} \frac{e(y-z)}{||y-z||} = 0$$

Choosing $z = x(t_0)$ and $y = x(t_0 + h)$ and dividing by $h \neq 0$

$$\frac{f(x(t_0+h)) - f(x(t_0))}{h}$$
$$= \sum_{i=1}^{L} \left[\frac{\partial f}{\partial x_i}(x(t_0)) \frac{(x_i(t_0+h) - x_i(t_0))}{h} \right] + \frac{e(x(t_0+h) - x(t_0))}{h}$$

Taking sequence $\{h_n\}$ converging to 0, we get:

$\lim_{h \to 0} \frac{f(x(t_0+h)) - f(x(t_0))}{h} = g'(t_0)$

$\lim_{h \to 0} \sum_{i=1}^{L} \left[\frac{\partial f}{\partial x_i}(x(t_0)) \frac{(x_i(t_0+h) - x_i(t_0))}{h} \right] = \nabla f(x(t_0)) \cdot x'(t_0)$

Proof: Chain Rule (Optional)

$$\lim_{h\to 0} \frac{e(x(t_0+h)-x(t_0))}{h}$$
$$= \lim_{h\to 0} \frac{e(x(t_0+h)-x(t_0))}{||x(t_0+h)-x(t_0)||} \cdot \lim_{h\to 0} \frac{||x(t_0+h)-x(t_0)||}{h}$$

The first term is 0 because , $\lim_{y\to z} \frac{e(y-z)}{||y-z||} = 0$.

The second term is

$$\lim_{h\to 0} \frac{||x(t_0+h)-x(t_0)||}{h}$$
$$= \lim_{h\to 0} \sqrt{\sum_{i=1}^{L} \left[\frac{x_i(t_0+h)-x_i(t_0)}{h}\right]^2}$$
$$= \sqrt{\sum_{i=1}^{L} \left[\lim_{h\to 0} \frac{x_i(t_0+h)-x_i(t_0)}{h}\right]^2} = \sqrt{\sum_{i=1}^{L} x_i'(t_0)^2} = ||x'(t_0)||$$

Thus, $\lim_{h\to 0} \frac{e(x(t_0+h)-x(t_0))}{h} = 0 \cdot ||x'(t_0)|| = 0$